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Transcriptomic profiling of *Paulownia elongata* in response to heat stress

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ABSTRACT

Rising global temperatures have caused a substantial loss in crop productivity throughout the world. Understanding the fundamental mechanisms underlying tolerance to heat stress in plants has become increasingly relevant, particularly as global warming continues to rise. The goal of our study was to identify and analyze heat stress-induced transcriptomic changes in *Paulownia elongata* plants. *Paulownia elongata* are fast growing trees known for their ability to grow in a wide range of temperatures. However, the genes responsible for its thermotolerance have not yet been identified and studied in detail. In our study, we used RNA-sequencing (RNA-seq) to analyze changes in gene expression at the transcriptomic level after inducing heat stress and identified 4,435 genes as differentially expressed. Most of the genes found to be upregulated play a critical role in protein binding, unfolding, retention, and transport, suggesting that heat tolerance in plants can be achieved by protecting native proteins. The genes and pathways identified in this study can serve as valuable targets in heat tolerance studies and development of heat tolerant plants. To the best of our knowledge, this is the first in-depth report of heat-induced transcriptome analysis in *P. elongata*.

1. Introduction

Life on Earth is currently being threatened by the rise in temperature, also called 'Global Warming' (Moernaut et al., 2018; Root et al., 2003). The most important loss due to global warming is happening in the agriculture industry including food and cash crops (Cline, 2007; Morton, 2007). Food availability is already at stake in nations around Sub-Saharan Africa and South Asia and is exacerbated by the rising temperatures in these regions (Sultan, 2012). Studies have reported that developing nations will be the first to be hit by the agricultural losses as they are closer to the equator, and an estimated 2% rise in temperature could cause a 25% loss in agricultural production (Aragón et al., 2018). This will affect the entire world as developing nations play a major role in global food production. Moreover, by 2080, global food demands are estimated to increase three-fold, which indicates that low, middle, and high-income countries need to analyze and adapt their agricultural practices before it is too late (Cline, 2007). This is where biotechnology can lend a helping hand by identifying the genes responsible for the desired phenotype (Ma et al., 2017; Bailey et al., 1996). In the current study, we focused on identifying genes that are necessary for plants to handle heat stress.

Plants, being sessile organisms, cannot move or hide away from the rising hot temperatures like we do. Therefore they undergo a lot of physical and metabolic changes that help them handle the heat stress (Hasanuzzaman et al., 2013). In response to severe heat, plants secrete more hormones and reactive oxygen species which in turn causes oxidative stress (Hasanuzzaman et al., 2013; Larkindale and Knight, 2002). This stress creates oxidative damage in plants and this will decrease their rate of survival (Larkindale and Knight, 2002). A recent study discovered different species of the *Paulownia* genus (Paulowniaceae family) are extremely resistant to temperature changes (Basu et al., 2016). All species of the *Paulownia* genus are reported to inherit this

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potential along with resistance to changes in soil composition. This is a major reason why they are seen all over the world. Besides the tolerance, *Paulownia* wood is also famous for their high cellulose content (Basu et al., 2016). Cellulose is the starting material for biofuel production and so *Paulownia* trees are regarded as a major source for biofuels (Basu et al., 2016; Sticklen, 2010). As biofuels are renewable and environmentally friendly, they can resolve the political uncertainty and environmental harms caused by overuse of petroleum (Sticklen, 2010). Thus it is vital to learn more about the vegetative tissues of *Paulownia* that contain cellulose for biofuel production (Basu et al., 2016). The vegetative tissues of plants are maintained and developed by factors involved in plant height, vegetative meristem activities, secondary wall biosynthesis, cell elongation and photosynthetic efficiency (Basu et al., 2016; Fernandez et al., 2009). Therefore a sound knowledge of the genes in *Paulownia* that have a major role in vegetative tissue growth is vital to improving tree breeding programs for biofuel production. The focus of this study is on *Paulownia elongata* as it is fast-growing and has immense potential to be used as a biofuel and bioethanol tree (Basu et al., 2016; Chaires et al., 2017). Though it is known that *P. elongata* is tolerant and adaptive to various temperatures (Chaires et al., 2017) the genes that are involved in handling the heat stress have not been characterized.

The main goal of this study was to understand the genome-wide gene expression patterns of *P. elongata* when subjected to heat stress. Identifying and analyzing the genes involved in handling heat stress will give us a broad idea about the molecular defense mechanisms available in plants to fight the threat of global warming. The data will also be useful in identifying the regulators of vegetative growth that act and protect the cellulosic biomass during hot temperatures. This will improve the tree breeding programs for biomass production and protect other crops from heat stress through the techniques of genetic engineering.

This RNA-seq based study investigated the differential gene expression of *P. elongata* when subjected to heat stress, which allowed for the quantification of expression levels of known and unknown genes during heat stress. The utilization of RNA-seq versus other techniques allows for a deeper understanding of these genes at the molecular level (Chaires et al., 2017; Mortazavi et al., 2008). Characterization of *P. elongata* heat stress-induced genes can pave the way to improving tree breeding and crop cultivation techniques for application in the fields of biofuel manufacturing and enhanced stress tolerance, respectively.

2. Methods

2.1. Plant Growth Conditions and Heat Stress Treatments

Six *Paulownia* plants that were three years old were used for the analysis. The plants were bought from *Paulownia* tree company, USA and grown for 14 days in Promix-HP Mycorrhizae potting soil (Constituents: Canadian Sphagnum peat moss (65%–70%), perlite and limestone (Premier Tech Horticulture, Canada). The plants were watered using regular tap water. In order to enrich the nutrient content of the soil, smart release-Osmocote plant fertilizer (Scotts Company, USA) was added to the soil. Three biological replicates were used for both control and heat stress-induced plants. The conditions for heat stress were 40°C for 24 hours with a photoperiod of 12 hours of light and 12 hours of dark achieved through a growth chamber. On the other hand, control plants were grown in a separate growth chamber at 25°C with the same photoperiod as that of the heat stressed plants. Heat stress was induced in *P. elongata* plants by placing them at 40°C for 24 hours from 8 AM. This condition was found to induce the maximum anti-oxidative activity in plants, which occurs as a response to tolerate heat stress (Hasanuzzaman et al., 2013).

2.2. RNA Sequencing

Total RNA was extracted from both control and treated *P. elongata* plants using Trizol reagent (Thermofisher Scientific, USA) following the manufacturer's instructions. The integrity of the RNA was confirmed by running an RNA gel using a Formaldehyde free RNA gel kit (Amresco, USA) following the manufacturer's protocol (Supplementary files 1 and 2). RNA from heat stressed plants was extracted from leaves after 24 hours of the heat stress treatment. Any genomic DNA contamination was removed with a DNase I treatment (Life Technologies, U.S.A). The remaining RNA was treated with the RNA clean and concentrator kit (Zymo Research, USA) using the manufacturer's protocol with modification (Ramadoss and Basu, 2018). After clean-up, cDNA libraries were synthesized (Supplementary file 2) and barcoded using the TrueSeq RNA Sample Prep Kit (v2; Illumina, U.S.A.) following the manufacturer's 'low sample protocol', and then pooled together for sequencing. Pair-end 76 bp reads were generated using the MiSeq Reagent Kit (v3; Illumina, U.S.A.) on an Illumina MiSeq instrument at the Department of Biology at California State University, Northridge. The total number of paired-end reads for each sample is listed in Table 1. The barcoded cDNA libraries were stored in a −20°C freezer.

2.3. Transcriptome Assembly and Gene Expression Analysis

To date, no public reference genome exists for *P. elongata*, therefore a *de novo* transcriptome was first generated by pooling all the Illumina sequencing reads and using Trinity software (v.2.0.6) with the default settings (Haas et al., 2013). The contigs were then examined for any potential contamination by BLAST searches against the NCBI non-redundant (nr) protein database with the blastx command in DIAMOND (v0.71) (Buchfink et al., 2015), and the output screened for non-plant hits using the MEGAN6 program (Huson et al., 2007). Contamination analysis identified 38 contigs as non-plant and were removed from the transcriptome. Completeness of the transcriptome was assessed by the "Completeness Assessment" functionality of OmicsBox (https://www.biobam.com/omicsbox; v1.3.11), which is based on Benchmarking Universal Single-Copy Orthologs (BUSCO, version 4.0.5; Seppey et al., 2019) and OrthoDB v10 (Kriventseva et al., 2015). The Viridiplantae lineage dataset was used with default BLAST parameters. Transcriptome sequences were annotated using Blast2GO (https://www.blast2go.com/) program.

Salmon (Patro et al., 2017) was used for quantification of reads, on the Cyverse computational infrastructure (Blast2GO), and gene expression analysis was conducted using the Bioconductor package, edgeR (Robinson et al., 2010). An exact Test was used to identify sequences with differential expression, and those with FDR < 0.05 and $|\log2FC| > 1$ were considered for downstream analysis. The identified DEGs were selected for GO and KEGG pathway analyses using OmicsBox. Blast2GO was used for functional annotation, Gene Ontology (GO) terms assignment, and GO enrichment analysis. The full set of annotated transcripts were used as the 'reference set'. The set of transcripts found to be upregulated or downregulated were used as the 'test set' to perform the GO enrichment analysis.

2.4. Data Availability

All data generated in this study have been registered, deposited, and publically available through the National Center for Biotechnology Information (NCBI). The project is registered under the BioProject

Table 1
Number of paired-end reads for each sample.

Biological samples	Reads
Heat stress sample 1 (H1)	7.9 million
Heat stress sample 2 (H2)	9.9 million
Heat stress sample 3 (H3)	6.8 million
Control sample 1 (C1)	6.4 million
Control sample 2 (C2)	6.2 million
Control sample 3 (C3)	2.6 million

accession number PRJNA485845 and BioSample accession number SUB4474744, whereas the raw sequence reads have been deposited in the Sequence Read Archive (SRA) under the accession number SRP158908. Gene expression (RNA-seq) data were deposited in the NCBI Gene Expression Omnibus (GEO) database under the accession number GSE119074. The transcriptome assembly is available as Supplementary file 3.

3. Results and Discussion

3.1. Heat Treatment

After induction of heat stress, mild phenotypic changes were observed through slightly drooping leaves. This phenological response was also recorded in other *Paulownia* species that were subjected to salt stress (Fan et al., 2016) and drought stress (Xu et al., 2014). The weakened water absorbing ability under the stress conditions possibly led to the drooping of leaves.

3.2. Transcriptome Assembly and Gene Expression Analysis

The assembled transcriptome consists of 100,186 contigs and is available as Supplementary file 3. Assessment of completeness of the transcriptome (Fig. 1a) shows that out of the 425 BUSCO groups used, 311 (73.18%) have complete gene representation (single-copy or duplicated), while 101 (23.76%) are partially recovered, and 13 (3.06%) are missing.

This set of 100,186 contigs was used as a reference database for gene expression analysis. 4,435 genes were identified to be differentially expressed under heat stress, using FDR < 0.05 and $|\log 2FC| > 1$ (Supplementary files 4 and 5). The number of genes that were upregulated and downregulated are 2,797 and 1,638 respectively. Blast2GO (Götz et al., 2008) software was used for functional annotation, resulting in functional annotation assignment for 71,529 contigs (Supplementary file 6). The top 25 differentially expressed genes (up and down regulated) are revealed in the heatmap with functional assignments (Fig. 1b).

3.3. Upregulated Genes

The RNA-seq data was used to hypothesize the potential roles of different genes in heat-stress mechanisms. The top 10 upregulated genes with their functional annotations have been listed in Table 2. The top upregulated gene found in this study was the gene coding for tonoplast dicarboxylate transporter (tDT). The tDT is a vacuolar dicarboxylate transporter which was the first malate carrier to be identified at the molecular level (Emmerlich et al., 2003). It transports malic acid from the vacuole to cytoplasm where it can be catabolized, leading to malic acid degradation (Etienne et al., 2013). The role of tDT in plant heat stress is not well studied but it has been reported in Arabidopsis that malate increased in concentration after an upsurge in temperature (Kaplan et al., 2004). Thus, we could hypothesize that tDT is also upregulated in conditions of heat stress to prevent malate accumulation or promote malate transportation. Further, tDT was also observed to be upregulated by overexpression of Jasmonate Responsive ERF 3 (JRE3) in transgenic tomatoes (Abdelkareem et al., 2019). JRE3 is a jasmonate responsive transcriptional factor that plays a major role in a wide range of plant responses from defense to abiotic stresses (Wasternack and Hause, 2013). JRE3 was also reported to be upregulated in the leaves of *Arabidopsis thaliana* grown under non-uniform salinity conditions (Zhang et al., 2019). Thus, tDT might be important in regulating the heat stress response of *P. elongata*.

The HSP 83–90 gene family includes heat shock proteins that were first identified in *A. thaliana*. It includes proteins of size 83–90 kilodalton. The names HSP83 and HSP90 are used interchangeably. HSP90 is an important factor in the regulation of heat stress response and small HSPs depend on them to promote thermotolerance (Duncan, 2005). HSP 90 promotes thermotolerance in Arabidopsis when recruited by heat-denatured polypeptides followed by subsequent activation of promoters of several heat inducible genes (Yamada et al., 2007). It was also reported to be induced in *Arabidopsis* leaves in response to certain viral infections (Whitham et al., 2006). HSP 83 is reported to be upregulated in response to aging, heat stress, hyperoxia, hydrogen peroxide, and ionizing radiation in *Drosophila melanogaster* (Landis et al., 2012). It is also known to play a major role in the heat stress response of moths (Gkouvitsas et al., 2009), fruit fly (Theodoraki and Mintzas, 2006), dust mites (Niu et al., 2020) and parasite Leishmania (Larreta et al., 2004). Thus HSP 83 might be playing a critical role in the thermotolerance of *P. elongata*.

The third most highly upregulated gene was identified as triacylglycerol (TAG) lipase which hydrolyzes the long chain triacylglycerol molecules (Beisson et al., 2000). This is a critical enzyme that provides the carbon skeletons and energy that promote growth after germination (Eastmond, 2006). While the role of TAG lipase during heat stress is not clear, it is known that specific lipases induce lipid remodeling as a result of abiotic stress. By doing this, they can reduce the formation of toxic lipid intermediates that cause membrane damage or cell death (Balogh et al., 2013). For example, in Arabidopsis, Heat-Inducible Lipase 1 (HIL1) is known to play a major role in the lipid

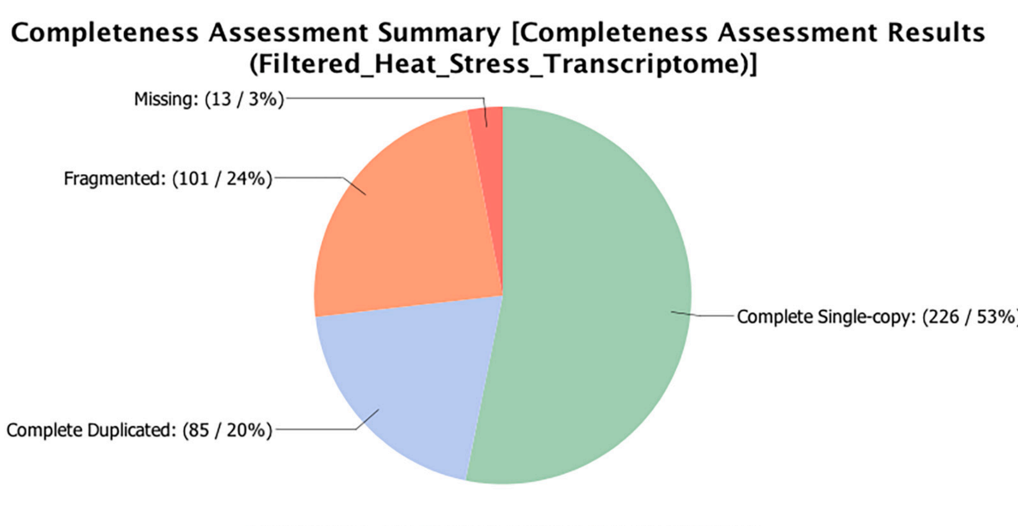

Fig. 1a. Transcriptome completeness assessment, using 425 BUSCO groups from the Viridiplantae lineage.

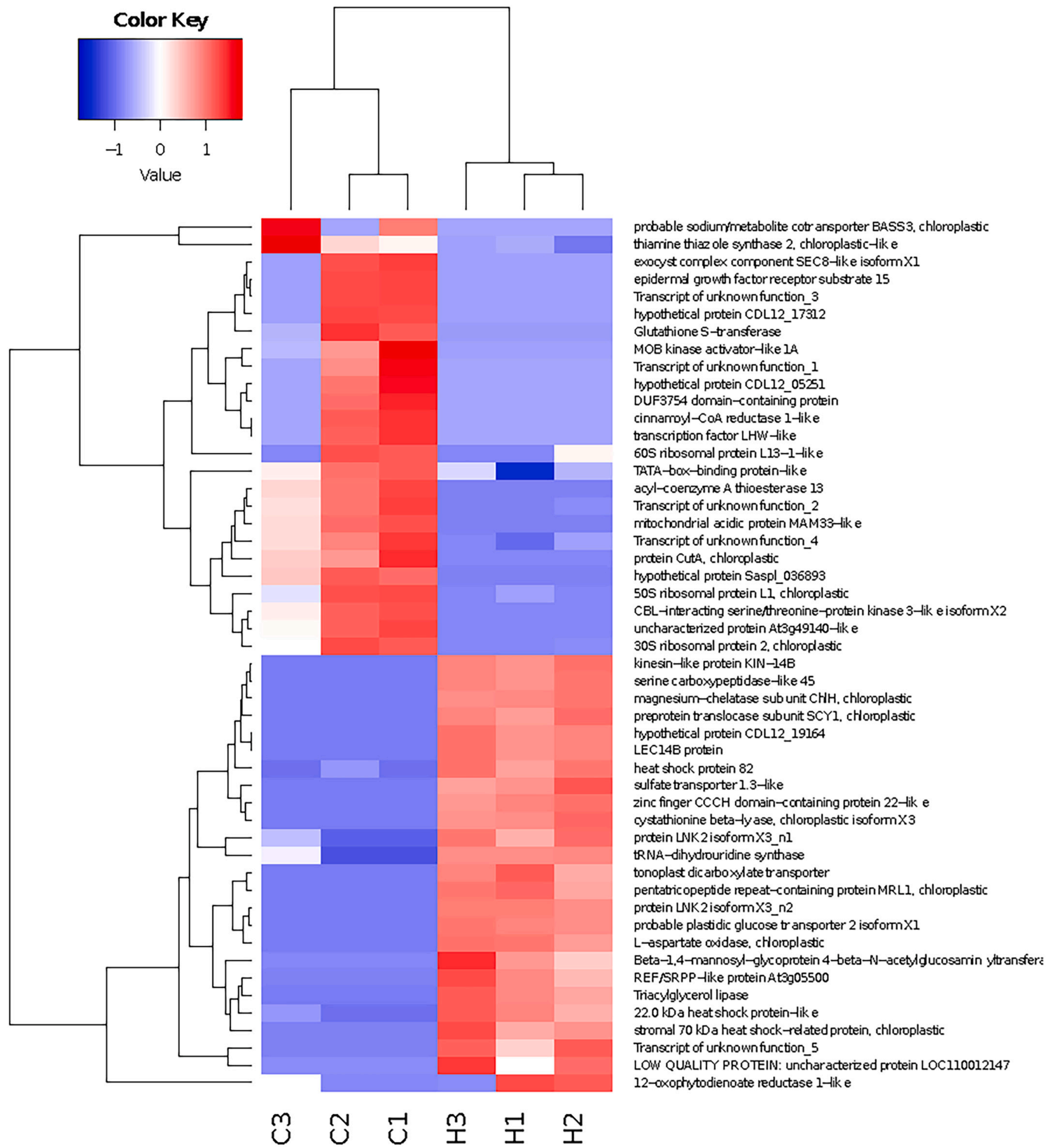

Fig. 1b. Heatmap showing expression of most differentially expressed genes (top 25 upregulated and downregulated) and log2 fold changes between control plants (C1, C2, C3) and plants exposed to heat stress (H1, H2, H3). Color corresponds to the per-gene z-score that is computed on logCPM (Counts per million mapped reads). Functional assignments for these transcripts are available in Supplementary file 7.

remodeling process induced by heat stress (Higashi et al., 2018). Thus, we can hypothesize that TAG lipase in *P. elongata* is involved in the heat stress response through lipid remodeling process.

Other genes that were upregulated include the cystathionine beta-lyase (CBL) which is a key enzyme of the methionine biosynthesis pathway, which has been reported to accumulate in plants during a combined stress condition of drought and heat (Koussevitzky et al., 2008). As methionine is a critical substrate used in the synthesis of various polyamines that are involved in stress tolerance (Alcázar et al., 2010), it can be justified why CBL is upregulated during heat stress. Next is beta-1,4-mannosyl-glycoprotein beta-1,4-N-acetylglucosaminyl-transferase which is involved in the synthesis of N- linked glycoprotein oligosaccharides. Although the role of this enzyme is not clear, some N-linked glycoproteins play vital roles in the abiotic stress response of

Table 2
Top 10 upregulated genes, sorted by log fold change (logFC: log fold change; logCPM: log counts per million; FDR: false discovery rate adjusted *p*-value).

Transcript_ID	Description	logFC	logCPM	*p*-Value	FDR
TR36547|c0_g1_i1	Tonoplast dicarboxylate transporter	11.2576634	5.7648584	6.87E-13	1.23E-08
TR20200|c0_g2_i1	Heat shock protein 82	10.7387723	7.46874261	1.33E-16	9.52E-12
TR54032|c0_g2_i1	Triacylglycerol lipase	10.2993624	4.82696562	4.95E-11	5.47E-07
TR2916|c0_g1_i2	Cystathionine beta-lyase, chloroplastic isoform X1	10.006529	4.54445103	2.19E-11	3.13E-07
TR70271|c0_g1_i9	Beta-1,4-mannosyl-glycoprotein 4-beta-N-acetylglucosaminyltransferase	9.98036445	4.52036512	2.49E-08	5.23E-05
TR13060|c0_g1_i1	L-aspartate oxidase, chloroplastic	9.93062768	4.47100833	5.35E-11	5.47E-07
TR269|c0_g3_i1	Zinc finger CCCH domain-containing protein 22-like	9.65205567	4.20830575	1.06E-10	8.44E-07
TR50586|c0_g1_i1	Stromal 70 kDa heat shock-related protein, chloroplastic-like	9.46926887	4.02875926	7.23E-09	1.85E-05
TR38143|c0_g1_i1	22.0 kDa heat shock protein-like	9.44679636	6.88854981	2.87E-13	1.03E-08
TR70304|c0_g1_i7	12-oxophytodienoate reductase 1-like	9.4308542	4.10608112	8.29E-06	0.0022377

plants (Zhang et al., 2009; Blanco-Herrera et al., 2015). Another significantly upregulated gene was ribulose bisphosphate carboxylase/oxygenase (Rubisco) activase. Rubisco activase is the key enzyme that regulates photosynthesis in plants. It functions by removing the inhibitors of Rubisco enzyme by ATP hydrolysis. This leads to the activation of Rubisco enzyme which is the rate limiting enzyme for the carbon fixation process (Kurek et al., 2007, Kumar et al. 2016). Moreover, it has been reported that Rubisco inactivation is common in heat-induced damage of higher plants like cotton, pea, tobacco, etc. (Kumar et al. 2016). Plants such as rice, wheat, and maize have been observed to overexpress Rubisco activase under heat stress (Kurek et al., 2007). Thus, plants protect their photosynthetic systems during heat stress by increasing the expression of Rubisco activase.

3.4. Downregulated Genes

We found 1,638 downregulated genes in heat stressed plants. The top 10 downregulated genes with their functional annotations are listed in Table 3. We identified 50S and 30S chloroplastic ribosomal genes as the two most highly downregulated genes. Photosynthesis is negatively affected in higher plants if chloroplasts are damaged due to heat stress. It is reported that plants can downregulate chloroplastic ribosomal protein genes which leads to transcriptional activation of heat responsive genes required for thermotolerance (Yu et al., 2012). Downregulation of chloroplastic genes was well documented by Lukoszek et al. (2016) through RNAseq experiments. Chloroplastic gene expression also controls the heat stress-mediated nuclear gene response (Yu et al., 2012). Cinnamoyl-CoA reductase (CCR) is an enzyme responsible for synthesis of lignin in plants (Ghosh et al., 2015). CCR was reported to be differentially expressed (both up and down regulated) in *Leucaena leucocephala* seedlings (Srivastava et al., 2015). Plants with downregulated CCR were reported to show developmental abnormalities along with reduced lignin content (Ghosh et al., 2015). Downregulation of CCR was reported to increase phenolic pools in tomato plants (*Solanum lycopersicum* L.) (Van Der Rest et al., 2006). We do not have any plausible explanation for the downregulation of CCR gene in paulownia plants. We speculated that the increased phenolic pool might protect plants from abiotic stress as reported by Sharma et al. (2019). The MAM33 gene family in plants is conserved and is found predominantly in the mitochondrial matrix (Jiang et al., 1999). The role of MAM33 in plants is yet to be elucidated. It is believed that MAM33 is involved in mitochondrial oxidative phosphorylation and nucleus–mitochondrion interactions (Jiang et al., 1999). In yeast, MAM33 is a critical matrix protein that ensures the proper assembly of ribosomes by binding to specific mito ribosomal proteins (Hillman and Henry, 2019). Moreover, this gene was also reported to be downregulated in cotton lines that possess cytoplasmic male sterility compared to fertile lines, which suggests that MAM33 may also be important in the male function of certain plants (Suzuki et al., 2013). Further studies are required to assess the role of MAM33 in a heat stress response.

Table 3
Top 10 downregulated genes, sorted by log fold change. (logFC: log fold change; logCPM: log counts per million; FDR: false discovery rate adjusted *p*-value).

Transcript_ID	Description	logFC	logCPM	*p*-Value	FDR
TR48060|c0_g1_i1	50S ribosomal protein L1, chloroplastic	-11.0609	5.5586	1.12E-06	0.0006
TR12208|c0_g1_i1	30S ribosomal protein 2, chloroplastic	-10.2687	4.7745	2.85E-06	0.00114
TR15090|c2_g1_i4	Cinnamoyl-CoA reductase 1-like	-10.1936	4.706	3.79E-06	0.0013
TR13504|c0_g1_i2	Hypothetical protein Saspl_036893	-10.0306	4.5435	9.62E-11	8.44E-07
TR65982|c0_g1_i2	Mitochondrial acidic protein MAM33-like	-9.98846	4.5044	9.98E-10	3.76E-06
TR41146|c0_g1_i1	Glutathione S-transferase	-9.94049	4.4497	1.02E-06	0.00056
TR4854|c0_g1_i2	Uncharacterized protein At3g49140-like	-9.59829	4.1238	8.23E-06	0.00223
TR66047|c0_g1_i1	DUF3754 domain-containing protein	-9.41295	3.9485	1.46E-05	0.0032
TR48144|c0_g1_i3	Epidermal growth factor receptor substrate 15	-9.14534	3.6819	2.03E-05	0.00393
TR36357|c0_g1_i4	CBL-interacting serine/threonine-protein kinase 3-like isoform X2	-9.00108	3.5477	1.92E-07	0.00018

3.5. Gene Ontology Analysis

Gene Ontology (GO) enrichment analysis of the upregulated genes shows that the top most categories by percentage of sequences are “ATP binding” and “chromosome organization” (Fig. 2). ATP binding is critical for the chaperone activity of Hsp70 proteins, which forms the central component of the cellular network of molecular chaperones and folding catalysts (Mayer and Bukau, 2005). Chromatin reorganization is reported to play a major role in plant defense to various stresses by changing nuclear structure, histone variants, chaperones, and modifications (Probst and Scheid, 2015). Most of the other upregulated genes are categorized into the “unfolded protein binding” GO term. Heat stress disrupts the protein folding in the endoplasmic reticulum (ER), thereby leading to ER stress (Schröder and Kaufman, 2005). This in turn elicits the unfolded protein response which is an evolutionarily conserved adaptive response to protect plants from heat stress. In maize, unfolded protein response is known to contribute to the nuclear/cytoplasmic heat shock response (Li et al., 2020). Enrichment of GO terms related to protein refolding, targeting, self-association, and misfolded protein binding suggests that *P. elongata* combats heat stress by reducing its impact on protein metabolism.

Among the upregulated genes, the most significantly enriched GO terms associated with molecular function were ‘unfolded protein

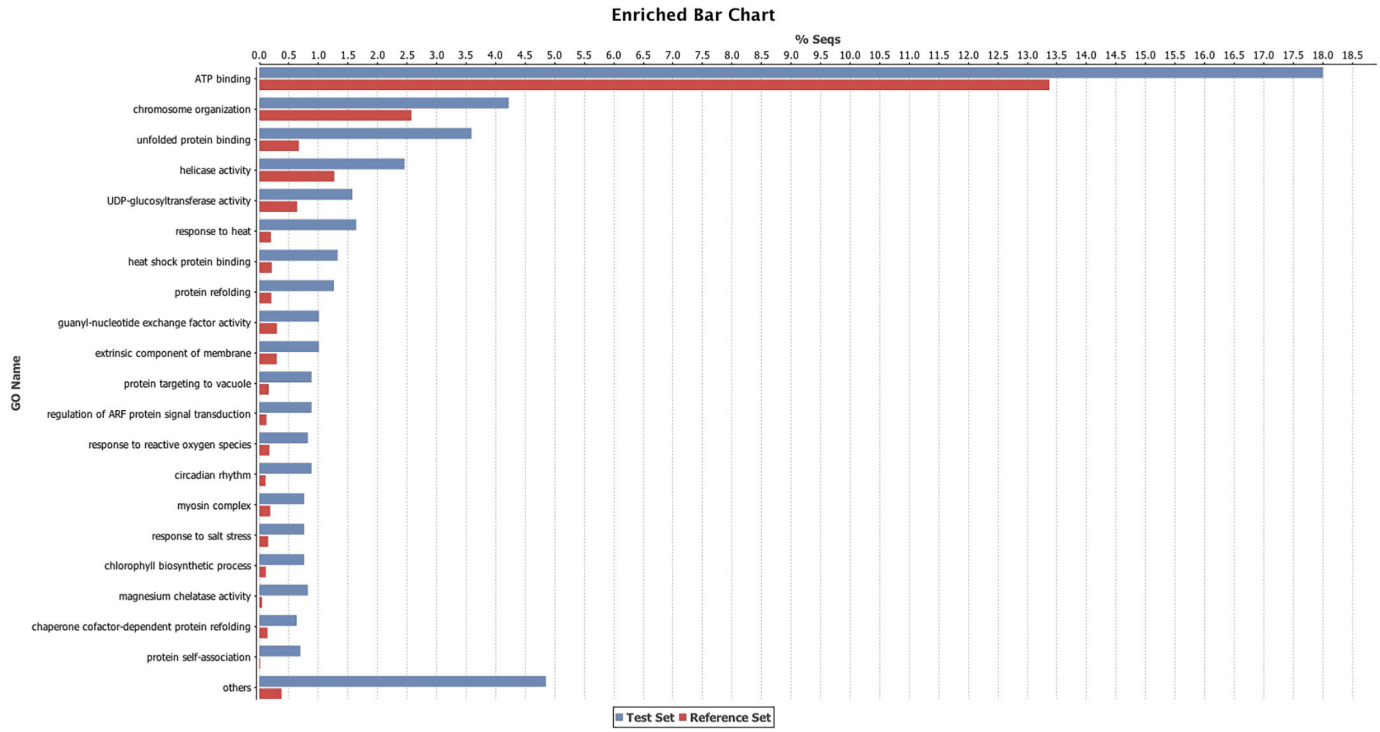

Fig. 2. Enriched GO terms based on percentage of sequences in the upregulated gene set.

binding', 'protein self-association', 'magnesium chelatase activity', and 'heat shock protein binding'. In terms of biological processes, 'response to heat', 'protein folding', and 'circadian rhythm', were the most significantly enriched terms (Supplementary files 8 and 13).

Gene Ontology (GO) enrichment analysis of the downregulated genes shows that the top most categories by percentage of sequences are "ribosome" and "structural constituent of ribosome" (Fig. 3). This is consistent with the study on genome-wide analysis of *Arabidopsis* subjected to heat shock which showed that ribosome biogenesis is sensitive to inhibition of translation (Yangueez et al., 2013). It was reported that mRNA translation of ribosomal genes was preferentially reduced during heat stress to save energy and curb the accumulation of misfolded proteins (Morimoto and Santoro, 1998; Yangueez et al., 2013). Similar inhibition in the translation of ribosomal structural components were also reported in mammalian cells as a result of ER stress (Ventoso et al., 2012). Other categories with a large number of downregulated genes

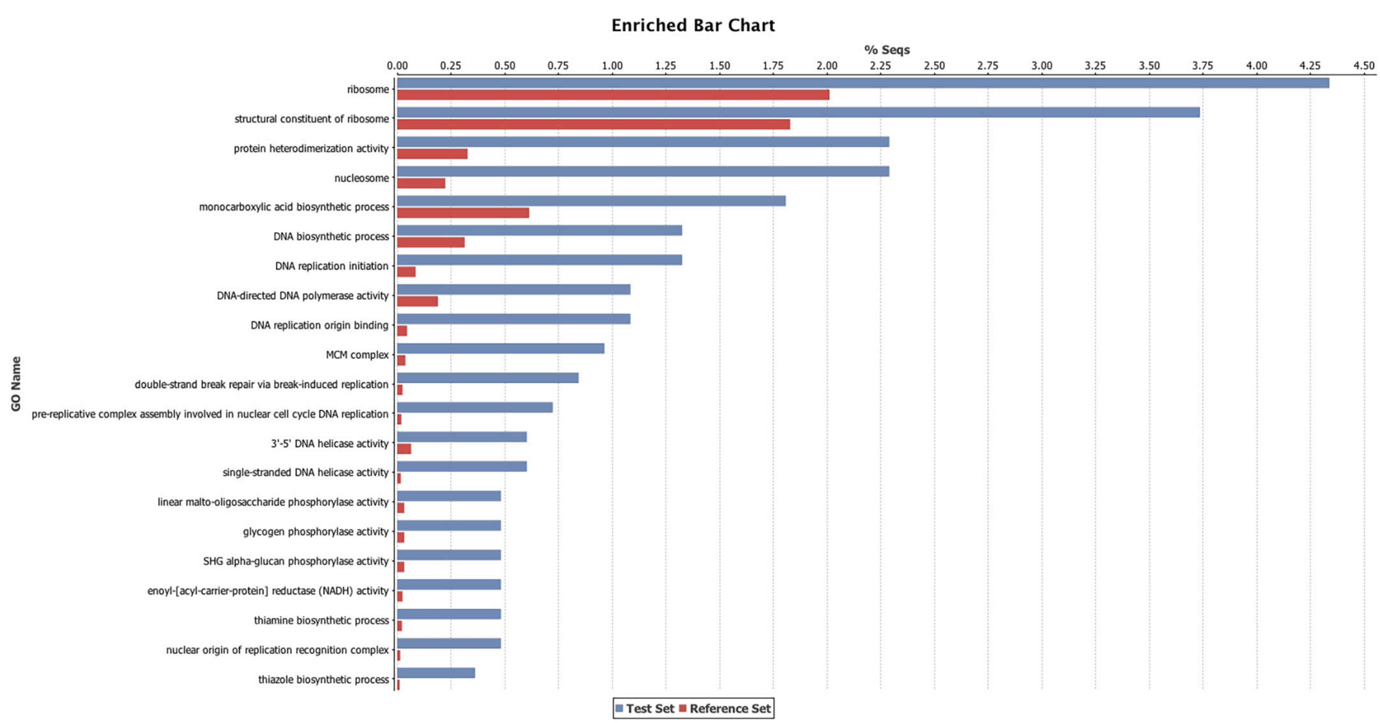

Fig. 3. Gene ontology enrichment analysis for downregulated genes.

were 'DNA replication', 'MCM complex', and related processes. Minichromosome Maintenance (MCM) proteins ensure accurate DNA replication in the S phase (Tuteja et al., 2011). Overall, the results suggest that *P. elongata* has defensive mechanisms that involve energy-saving and protection of proteins. We predict the downregulation of the fundamental DNA replication processes might affect plant growth during extreme heat stress. Further sequencing and annotation of the *P. elongata* genome would help characterize the transcriptome and understand heat stress-related pathways better.

Among the downregulated genes, the most significantly enriched GO terms associated with molecular function were 'protein heterodimerization activity', and 'DNA replication origin binding'. In terms of biological processes, 'DNA replication initiation' and related terms were the most significantly enriched (Supplementary files 9 and 14).

3.6. Biological Processes Enriched in *Paulownia elongata* Plants

Biological Processes enriched as a result of heat stress are shown in Supplementary file 10. The top enriched biological processes that involved a response to stimulus in *P. elongata* were responses to heat stress, salt stress, and reactive oxygen species (ROS). As mentioned above in the GO enrichment analysis, there were genes upregulated in the study that are commonly transcribed in response to both heat and salt stress. Heat stress in general allows enhanced transpiration to cool the plant which could be detrimental when combined with salinity stress as that allows more salt uptake (Mittler, 2006). However, some studies have shown that heat and salt stress responses can be complementary in plants and therefore render a higher degree of stress tolerance (Rivero et al., 2013; Wen et al., 2005). One of the key factors involved in improved heat and salt tolerance is the biosynthesis of trehalose (Rivero et al., 2013) which was observed to be enriched in our heat stressed plants. It has been shown that a variety of abiotic stresses such as heat, salt, cold, drought are all capable of accumulating reactive oxygen species causing impairment of cellular processes and photosynthesis (Sunkar et al., 2003). Thus, an enriched response to ROS is again complementary to the thermotolerance of *P. elongata*. As *P. elongata* is already reported to be salt-tolerant (Chaires et al., 2017), we hypothesize that a combined response to heat, salt, and ROS are helping the plant in improving its degree of tolerance.

Another important resistance factor that is enriched in *P. elongata* and is involved in a wide range of environmental stresses such as drought, osmotic and heavy metal contamination is the biosynthesis of tetrapyrroles (Phung et al., 2011; Cui et al., 2013; Zhang et al., 2015). Tetrapyrroles are a group of compounds that include chlorophyll, heme, and other precursors that have biologically important functions (Tanaka and Tanaka, 2007). This suggests that *P. elongata* might not just be tolerant of heat and salt but also to several other abiotic stresses.

Other enriched biological pathways in response to heat stress in *P. elongata* involve protein folding, protein binding, retention, and vacuolar transport. Generally, heat shock proteins or chaperones are the

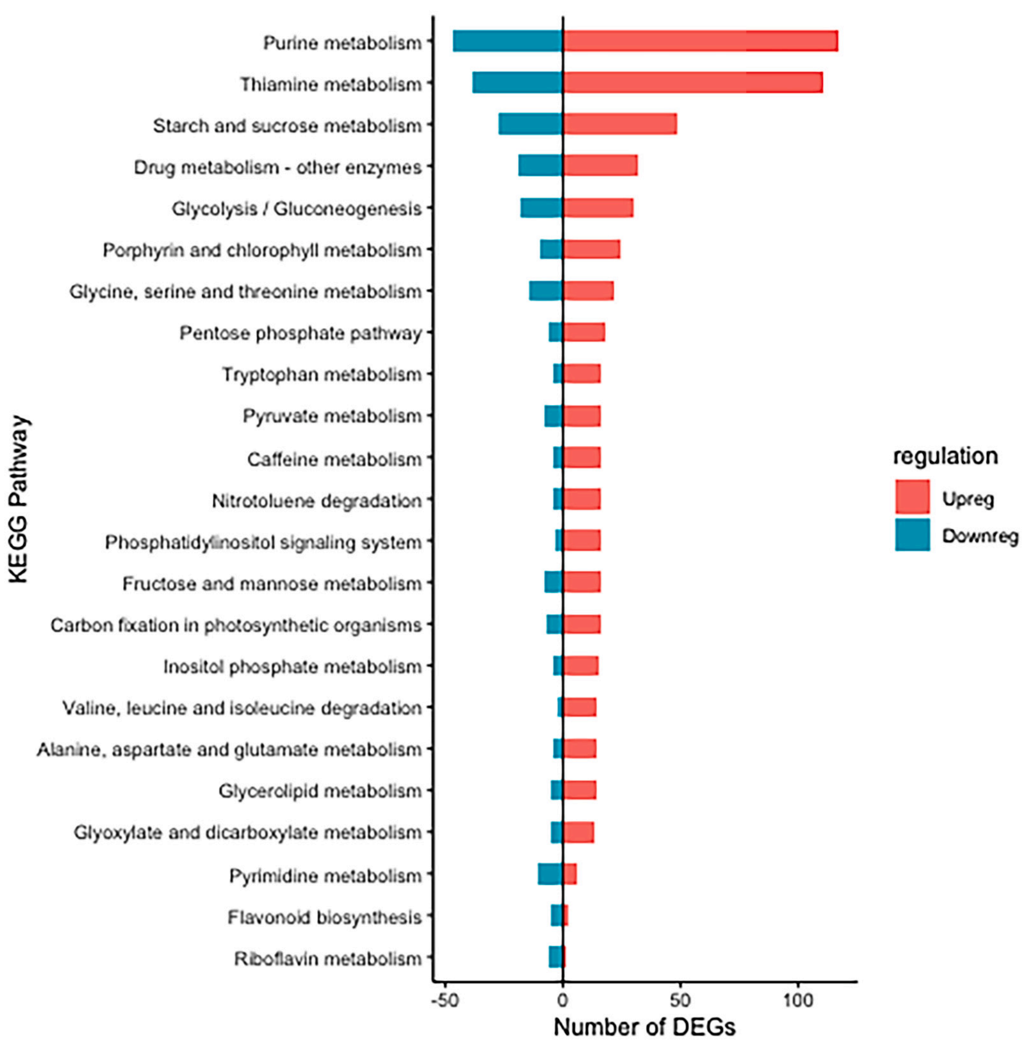

Fig. 4. Top KEGG pathways enriched for up and downregulated genes for heat stress.

ones that are actively involved in all the above-mentioned functions in response to high heat (Hartl et al., 2011; Benesova et al., 2012; Balchin et al., 2016). As several heat shock proteins were upregulated in our analyses such as HSP83, HSP70, 22 kDa Class IV HSP, it could be justified that these protein-related processes are being regulated by those genes. Recently, HSPs were found to be positively interacting with defense proteins thereby regulating biotic stress tolerance in potato plants (Yogendra et al., 2015). Functional analyses on heat shock proteins of *P. elongata* can reveal if their HSPs are capable of providing biotic stress tolerance to them. In summary, *P. elongata* plants are able to handle heat stress by enhancing their stimulus processes that are applicable to a wide range of stresses.

3.7. Pathway Analysis of Differentially Expressed Genes

KEGG pathway analysis of the DEGs revealed that 'Purine metabolism', 'Thiamine metabolism', 'Starch and sucrose metabolism', 'Glycolysis/Glucaneogenesis', and 'Porphyrin and chlorophyll metabolism', were the most active pathways (Fig. 4). The upregulated genes in *P. elongata* were identified to be involved in glycolysis/gluconeogenesis, glyoxylate and dicarboxylate metabolism, glutathione metabolism, tryptophan metabolism, pyruvate metabolism, cyanoamino acid metabolism, photosynthesis, biosynthesis of antibiotics and caffeine metabolism (See Supplementary file 11, upregulated genes in KEGG pathways). As temperature rises, plants respire more (Downs and Heckathorn, 1998). Through respiratory pathways, such as glycolysis, the plants maintain their energy production and several physiological functions (Fernie et al. 2004). Although the photosynthesis rate declines with temperature, the upregulated genes, biological processes and pathways related to photosynthesis were enriched showing that *P. elongata*'s photosynthesis is not affected by heat stress. The antibiotics biosynthesis pathway upregulation is consistent with the heat stress response of several microorganisms (Doull et al.1993; Feng et al. 2019) and a fungi, *Lentinula edodes*, suggesting that antibiotics play a major role in thermotolerance (Wang et al. 2018). Other enriched pathways are associated with amino acid metabolism which is closely related to abiotic stress tolerance of plants (Zörb et al., 2004; Haitao et al., 2013, Zhang et al. 2017). The downregulated genes in *P. elongata* have been identified to be mostly related to cell cycle pathways such as DNA replication initiation, DNA replication origin binding, nucleosome, MCM complex, and double stranded break repair. (See Supplementary file 12, downregulated genes in KEGG pathways). This could be seen as an adaptive response because plants temporarily arrest the cell cycle to avoid consequences of stress damage (Takahashi et al., 2019). Moreover, a recent study on Arabidopsis showed that nucleosome "H2A.Z" levels were reduced during temperature stress, so as to activate the signals for heat stress responsive genes (Cortijo et al., 2017). Since this analysis is exploratory in nature, our findings warrant further investigation.

4. Conclusion

Global warming is negatively affecting crop productivity around the world which propels the need to understand heat stress tolerance in plants. Paulownia trees are found growing over a wide range of temperatures from 15 to 33°C. However, the genes and pathways that are involved in *P. elongata*'s adaptation to a wide range of temperatures have not been extensively studied. This study identified 4,435 genes in *P. elongata* that were differentially expressed under heat stress using RNA-seq analysis. The results from this study revealed that the top genes upregulated were linked to not only heat tolerance but also to other abiotic stresses, particularly salt tolerance. Most of the heat tolerant genes were coding for a particular class of proteins called HSP – heat shock proteins. The top enriched pathways of heat stress response in *P. elongata* involved protein folding, protein binding, protein retention, vacuolar transport, and circadian rhythm. HSPs are known to protect the plants by controlling the vast protein denaturation that occurs as a consequence of heat stress. Thus it is predicted that *P. elongata* tolerates heat stress by timed stress responses that help protect their proteins from the consequences of heat stress. The genes identified in the study can serve as potential targets that need to be differentially expressed in order to enhance heat tolerance in agricultural crops. Further, the proposed genes can also be used in the engineering of *P. elongata* plants for the production of second-generation biofuels. Functional characterization of these differentially expressed genes in the future is required to improve our understanding of their synergistic action against heat stress.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.plgene.2021.100330.

Declaration of Competing Interest

The authors report no relationships that could be construed as a conflict of interest.

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References

Abdelkareem, A., Thagun, C., Imanishi, S., Hashimoto, T., Shoji, T., 2019. Identification of genes regulated by a jasmonate-and salt-inducible transcription factor JRE3 in tomato. Plant Biotechnol. 36 (1), 29–37.

Alcázar, R., Altabella, T., Marco, F., et al., 2010. Polyamines: molecules with regulatory functions in plant abiotic stress tolerance. Planta 231, 1237–1249.

Aragón, F.M., Oteiza, F., Rud, J.B., 2018. Climate Change and Agriculture: Farmer Adaptation to Extreme Heat. Discussion Papers dp18-02. Department of Economics, Simon Fraser University.

Bailey, J.E., Sburlati, A., Hatzimanikatis, V., Lee, K., Renner, W.A., Tsai, P.S., 1996. Inverse metabolic engineering: a strategy for directed genetic engineering of useful phenotypes. Biotechnol. Bioeng. 52 (1), 109–121.

Balchin, D., Hayer-Hartl, M., Hartl, F.U., 2016. In vivo aspects of protein folding and quality control. Science 353, acc4353.

Balogh, G., Péter, M., Glatz, A., et al., 2013. Key role of lipids in heat stress management. FEBS Lett. 587, 1970–1980.

Basu, C., Joshee, N., Gezalian, T., Vaidya, B.N., Satidkit, A., Hemmati, H., Perry, Z.D., 2016. Cross-species PCR and field studies on *Paulownia elongata*: a potential bioenergy crop. Bioethanol 2 (1), 12–23.

Beisson, F., Tiss, A., Rivière, C., Verger, R., 2000. Methods for lipase detection and assay: a critical review. Eur. J. Lipid Sci. Technol. 102, 133–153.

Benesova, M., Hola, D., Fischer, L., Jedelsky, P.L., Hnilicka, F., Wilhelmova, N., Rothova, O., Kocova, M., Prochazkova, D., Honnerova, J., 2012. The physiology and proteomics of drought tolerance in maize: early stomatal closure as a cause of lower tolerance to short-term dehydration. PLoS One 7, e38017.

Blanco-Herrera, F., Moreno, A.A., Tapia, R., et al., 2015. The UDP-glucose: glycoprotein glucosyltransferase (UGGT), a key enzyme in ER quality control, plays a significant role in plant growth as well as biotic and abiotic stress in *Arabidopsis thaliana*. BMC Plant Biol. 15 (1), 127.

Buchfink, B., Xie, C., Huson, D.H., 2015. Fast and sensitive protein alignment using DIAMOND. Nat. Methods 12, 59–60.

Chaires, M., Gupta, D., Joshee, N., Cooper, K.K., Basu, C., 2017. RNA-seq analysis of the salt stress-induced transcripts in fast-growing bioenergy tree, Paulownia elongata. J. Plant Interact. 12 (1), 128–136.

Cline, W.R., 2007. Global warming and agriculture: impact estimates by country. Center for Global Development, Washington, D.C.

Cortijo, S., Charoensawan, V., Brestovitsky, A., Buning, R., Ravarani, C., Rhodes, D., van Noort, J., Jaeger, K.E., Wigge, P.A., 2017. Transcriptional Regulation of the Ambient Temperature Response by H2A.Z Nucleosomes and HSF1 Transcription Factors in Arabidopsis. Mol. Plant 10 (10), 1258–1273. https://doi.org/10.1016/j.molp.2017.08.014.

Cui, W., Zhang, J., Xuan, W., Xie, Y., 2013. Up-regulation of heme oxygenase-1 contributes to the amelioration of aluminum-induced oxidative stress in *Medicago sativa*. J. Plant Physiol. 170, 1328–1336.

Duncan, R., 2005. Inhibition of Hsp90 function delays and impairs recovery from heat shock. FEBS J. 272, 5244–5256.

Eastmond, P.J., 2006. SUGAR-DEPENDENT1 encodes a patatin domain triacylglycerol lipase that initiates storage oil breakdown in germinating Arabidopsis seeds. Plant Cell 18 (3), 665–675.

Emmerlich, V., Linka, N., Reinhold, T., et al., 2003. The plant homolog to the human sodium/dicarboxylic cotransporter is the vacuolar malate carrier. Proc. Natl. Acad. Sci. 100, 11122–11126.

Etienne, A., Génard, M., Lobit, P., et al., 2013. What controls fleshy fruit acidity? A review of malate and citrate accumulation in fruit cells. J. Exp. Bot. 64, 1451–1469.

Fan, G., Wang, L., Deng, M., Zhao, Z., Dong, Y., Zhang, X., Li, Y., 2016. Changes in transcript related to osmosis and intracellular ion homeostasis in *Paulownia tomentosa* under salt stress. Front. Plant Sci. 7, 384.
Fernandez, M.G.S., Becraft, P.W., Yin, Y., Lübberstedt, T., 2009. From dwarves to giants? Plant height manipulation for biomass yield. Trends Plant Sci. 14, 454–461.
Ghosh, Ritesh, Choi, Bosung, Kim, Jonggeun, Jeong, Mi-Jeong, Mohanta, Tapan Kumar, Bae, Hanhong, 2015. Developmental- and stress-mediated expression analysis of cinnamoyl-CoA reductase 1 (CCR1) from *Hibiscus cannabinus*. J. Plant Interact. 10 (1), 158–166.
Gkouvitsas, T., Kontogiannatos, D., Kourti, A., 2009. Expression of the Hsp83 gene in response to diapause and thermal stress in the moth Sesamia nonagrioides. Insect Mol. Biol. 18 (6), 759–768.
Götz, S., García-Gómez, J.M., Terol, J., Williams, T.D., et al., 2008. High-throughput functional annotation and data mining with the Blast2GO suite. Nucleic Acids Res. 36 (10), 3420–3435.
Haas, B.J., Papanicolaou, A., Yassour, M., Grabherr, M., Blood, P.D., Bowden, J., Couger, M.B., Eccles, D., Li, B., Lieber, M., et al., 2013. De novo transcript sequence reconstruction from RNA-seq using the trinity platform for reference generation and analysis. Nat. Protoc. 8, 1494–1512.
Hartl, F.U., Bracher, A., Hayer-Hartl, M., 2011. Molecular c Haperones in Protein Folding and Proteostasis. Nature. 475, 324–332.
Hasanuzzaman, M., Nahar, K., Alam, M., Roychowdhury, R., Fujita, M., 2013. Physiological, biochemical, and molecular mechanisms of heat stress tolerance in plants. Int. J. Mol. Sci. 14 (5), 9643–9684.
Higashi, Y., Okazaki, Y., Takano, K., et al., 2018. Heat inducible lipase1 remodels chloroplastic monogalactosyldiacylglycerol by liberating α-linolenic acid in arabidopsis leaves under heat stress. Plant Cell 30, 1887–1905.
Hillman, G.A., Henry, M.F., 2019. The yeast protein Mam33 functions in the assembly of the mitochondrial ribosome. J. Biol. Chem. 294 (25), 9813–9829.
Huson, D.H., Auch, A.F., Qi, J., Schuster, S.C., 2007. Megan analysis of metagenomic data. Genome Res. 17, 377–386.
Jiang, J., Zhang, Y., Krainer, A., Xu, R.-M., 1999. Crystal structure of human P32, a doughnut-shaped acidic mitochondrial matrix protein. Proc. Natl. Acad. Sci. 96 (7), 3572–3577.
Kaplan, F., Kopka, J., Haskell, D.W., et al., 2004. Exploring the temperature-stress metabolome of Arabidopsis. Plant Physiol. 136 (4), 4159–4168.
Koussevitzky, S., Suzuki, N., Huntington, S., et al., 2008. Ascorbate peroxidase 1 plays a key role in the response of arabidopsis thaliana to stress combination. J. Biol. Chem. 283, 34197–34203.
Kriventseva, E.V., Tegenfeldt, F., Petty, T.J., Waterhouse, R.M., Simao, F.A., Pozdnyakov, I.A., Ioannidis, P., Zdobnov, E.M., NAR, 2015. OrthoDB v8: update of the hierarchical catalog of orthologs and the underlying free software (Jan).
Kurek, I., Chang, T.K., Bertain, S.M., Madrigal, A., Liu, L., Lassner, M.W., Zhu, G., 2007. Enhanced thermostability of *Arabidopsis rubisco* activase improves photosynthesis and growth rates under moderate heat stress. Plant Cell 19 (10), 3230–3241.
Landis, G., Shen, J., Tower, J., 2012. Gene expression changes in response to aging compared to heat stress, oxidative stress and ionizing radiation in *Drosophila melanogaster*. Aging 4 (11), 768–789.
Larkindale, J., Knight, M.R., 2002. Protection against heat stress-induced oxidative damage in arabidopsis involves calcium, abscisic acid, ethylene, and salicylic acid. Plant Physiol. 128 (2), 682–695.
Larreta, R., Soto, M., Quijada, L., et al., 2004. The expression of HSP83 genes in Leishmania infantum is affected by temperature and by stage-differentiation and is regulated at the levels of mRNA stability and translation. BMC Mol. Biol. 5 (1), 3.
Li, Z., Tang, J., Srivastava, R., Bassham, D.C., Howell, S.H., 2020. The transcription factor bZIP60 links the unfolded protein response (UPR) to the heat stress response (HSR) in maize. Plant Cell 32 (11), 3559–3575.
Lukoszek, R., Feist, P., Ignatova, Z., 2016. Insights into the adaptive response of *Arabidopsis thaliana* to prolonged thermal stress by ribosomal profiling and RNA-Seq. BMC Plant Biol. 16, 221.
Ma, J., Gilmour, B., Dang, H., 2017. Promise, problems and prospects: agri-biotech governance in China, India and Japan. China Agric. Econ. Rev. 9 (3), 453–475.
Mayer, M.P., Bukau, B., 2005. Hsp70 chaperones: cellular functions and molecular mechanism. Cell. Mol. Life Sci. 62, 670.
Mittler, R., 2006. Abiotic stress, the field environment and stress combination. Trends Plant Sci. 11, 15–19.
Moernaut, R., Mast, J., Pauwels, L., 2018. Framing climate change: a multi-level model. In: Leal Filho, W., Manolas, E., Azul, A., Azeiteiro, U., McGhie, H. (Eds.), Handbook of Climate Change Communication, Climate Change Management, 1. Springer, Cham, Switzerland, pp. 215–271.
Morimoto, R.I., Santoro, M.G., 1998. Stress-inducible responses and heat shock proteins: new pharmacologic targets for cytoprotection. Nat. Biotechnol. 16, 833–838.
Mortazavi, A., Williams, B.A., McCue, K., Schaeffer, L., Wold, B., 2008. Mapping and quantifying mammalian transcriptomes by RNA-Seq. Nat. Methods 5 (7), 621–628.
Morton, J.F., 2007. The impact of climate change on smallholder and subsistence agriculture. Proc. Natl. Acad. Sci. 104 (50), 19680–19685.
Niu, D., Zhao, Y., Gong, X., et al., 2020. Stress response and silencing verification of heat shock proteins in Dermatophagoides farinae under temperature stress. Int. J. Biol. Macromol. 144, 351–361.
Patro, R., Duggal, G., Love, M.I., Irizarry, R.A., Kingsford, C., 2017. Salmon provides fast and bias-aware quantification of transcript expression. Nature Methods. Philos. Trans. Roy. Soc. B: Biol. Sci. 365 (1554), 2973–2989.
Phung, T.H., Jung, H.I., Park, J.H., Kim, J.G., Back, K., Jung, S., 2011. Porphyrin biosynthesis control under water stress: sustained porphyrin status correlates with drought tolerance in transgenic rice. Plant Physiol. 157, 1746–1764.
Probst, A.V., Scheid, O.M., 2015. Stress-induced structural changes in plant chromatin. Curr. Opin. Plant Biol. 27, 8–16.
Ramadoss, N., Basu, C., 2018. Extraction of RNA from recalcitrant tree species *Paulownia elongata*. Bio-protocol 8 (14), e2925.
Rivero, R.M., Mestre, T.C., Mittler, R.O.N., Rubio, F., Garcia-Sanchez, F., Martinez, V., 2013. The combined effect of salinity and heat reveals a specific physiological, biochemical and molecular response in tomato plants. Plant Cell Environ. 37 (5), 1059–1073.
Robinson, M.D., McCarthy, D.J., Smyth, G.K., 2010. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. Bioinformatics 26 (1), 139–140.
Root, T.L., Price, J.T., Hall, K.R., Schneider, S.H., Rosenzweig, C., Pounds, J.A., 2003. Fingerprints of global warming on wild animals and plants. Nature 421 (6918), 57–60.
Schröder, M., Kaufman, R.J., 2005. The mammalian unfolded protein response. Annu. Rev. Biochem. 74, 739–789.
Seppey, M., Manni, M., Zdobnov, E.M., 2019. BUSCO: assessing genome assembly and annotation completeness. In: Kollmar, M. (Ed.), Gene Prediction. Methods in Molecular Biology, vol 1962. Humana, New York, NY.
Sharma, A., Shahzad, B., Rehman, A., Bhardwaj, R., Landi, M., Zheng, B., 2019. Response of phenylpropanoid pathway and the role of polyphenols in plants under abiotic stress. Molecules 24 (13), 2452.
Srivastava, S., Vishwakarma, R.K., Arafat, Y.A., Gupta, S.K., Khan, B.M., 2015 Apr. Abiotic stress induces change in Cinnamoyl CoA Reductase (CCR) protein abundance and lignin deposition in developing seedlings of *Leucaena leucocephala*. Physiol. Mol. Biol. Plants 21 (2), 197–205.
Sticklen, M.B., 2010. Retraction note to: plant genetic engineering for biofuel production: towards affordable cellulosic ethanol. Nat. Rev. Genet. 11 (4), 308.
Sultan, B., 2012. Global warming threatens agricultural productivity in Africa and South Asia. Environ. Res. Lett. 7 (4), 041001.
Sunkar, R., Bartels, D., Kirch, H.H., 2003. Overexpression of a stress-inducible aldehyde dehydrogenase gene from *Arabidopsis thaliana* in transgenic plants improves stress tolerance. Plant J. 35, 452–464. https://doi.org/10.1046/j.1365-313X.2003.01819.x.
Suzuki, H., Rodriguez-Uribe, L., Xu, J., Zhang, J., 2013. Transcriptome analysis of cytoplasmic male sterility and restoration in CMS-D8 cotton. Plant Cell Rep. 32 (10), 1531–1542.
Takahashi, N., Ogita, N., Takahashi, T., Taniguchi, S., Tanaka, M., Seki, M., Umeda, M., 2019. A regulatory module controlling stress-induced cell cycle arrest in Arabidopsis. Elife 8, e43944.
Tanaka, R., Tanaka, A., 2007. Tetrapyrrole biosynthesis in higher plants. Annu. Rev. Plant Biol. 58, 321–346.
Theodoraki, M.A., Mintzas, A.C., 2006. cDNA cloning, heat shock regulation and developmental expression of the hsp83 gene in the Mediterranean fruit fly *Ceratitis capitata*. Insect Mol. Biol. 15 (6), 839–852.
Tuteja, N., Tran, N.Q., Dang, H.Q., et al., 2011. Plant MCM proteins: role in DNA replication and beyond. Plant Mol. Biol. 77, 537–545.
Van Der Rest, B., Danoun, S., Boudet, A.M., Rochange, S.F., 2006. Down-regulation of cinnamoyl-CoA reductase in tomato (*Solanum lycopersicum* L.) induces dramatic changes in soluble phenolic pools. J. Exp. Bot. 57, 1399–1411.
Ventoso, I., Kochetov, A., Montaner, D., Dopazo, J., Santoyo, J., 2012. Extensive translatome remodeling during ER stress response in mammalian cells. PLoS One 7.
Wasternack, C., Hause, B., 2013. Jasmonates: Biosynthesis, perception, signal transduction and action in plant stress response, growth and development. An update to the 2007 review in Annals of Botany. Ann. Bot. 111, 1021–1058.
Wen, X., Qiu, N., Lu, Q., Lu, C., 2005. Enhanced thermotolerance of photosystem II in salt-adapted plants of the halophyte *Artemisia anethifolia*. Planta 220 (3), 486–497.
Whitham, S.A., Yang, C., Goodin, M.M., 2006. Global impact: elucidating plant responses to viral infection. Mol. Plant-Microbe Interact. 19 (11), 1207–1215.
Xu, E., Fan, G., Niu, S., Zhao, Z., Deng, M., Dong, Y., 2014. Transcriptome-wide profiling and expression analysis of diploid and autotetraploid *Paulownia tomentosa×Paulownia fortunei* under drought stress. PLoS One 9 (11), e113313.
Yamada, K., Fukao, Y., Hayashi, M., et al., 2007. Cytosolic HSP90 regulates the heat shock response that is responsible for heat acclimation in *Arabidopsis thaliana*. J. Biol. Chem. 282, 37794–37804.
Yangueez, E., Castro-Sanz, A.B., Fernandez-Bautista, N., Oliveros, J.C., Castellano, M.M., 2013. Analysis of genome-wide changes in the translatome of Arabidopsis seedlings subjected to heat stress. PLoS One 8 (8), e71425.
Yogendra, K.N., Kumar, A., Sarkar, K., Li, Y., Pushpa, D., Mosa, K.A., Kushalappa, A.C., 2015. Transcription factor StWRKY1 regulates phenylpropanoid metabolites conferring late blight resistance in potato. J. Exp. Bot. 66 (22), 7377–7389.
Yu, H.-D., Yang, X.-F., Chen, S.-T., Wang, Y.-T., Li, J.-K., Shen, Q., et al., 2012. Downregulation of chloroplast RPS1 negatively modulates nuclear heat-responsive expression of *HsfA2* and its target genes in Arabidopsis. PLoS Genet. 8 (5), e1002669.
Zhang, H., Wang, R., Wang, H., et al., 2019. Heterogeneous root zone salinity mitigates salt injury to *Sorghum bicolor* (L.) Moench in a split-root system. PLoS One 14 (12), e0227020.
Zhang, M., Henquet, M., Chen, Z., et al., 2009. LEW3, encoding a putative α-1, 2-mannosyltransferase (ALG11) in N-linked glycoprotein, plays vital roles in cell-wall biosynthesis and the abiotic stress response in *Arabidopsis thaliana*. Plant J. 60 (6), 983–999.
Zhang, Z., Zhang, G., Zhu, F., et al., 2015. The roles of tetrapyrroles in plastid retrograde signaling and tolerance to environmental stresses. Planta 242, 1263–1276.